Machine Learning for Anomaly and Attack Detection

MSI – MEI – MI – MDS
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INTOL

Anomaly Detection

- Anomalies are event patterns that are different from a certain (reference) expected behavior
- Anomaly detection techniques are typically employed to
  - <u>Detect intrusions</u>
  - Detect credit card frauds
  - System failures
  - Detect anomalies in medical and public health records
  - Image processing
  - ...

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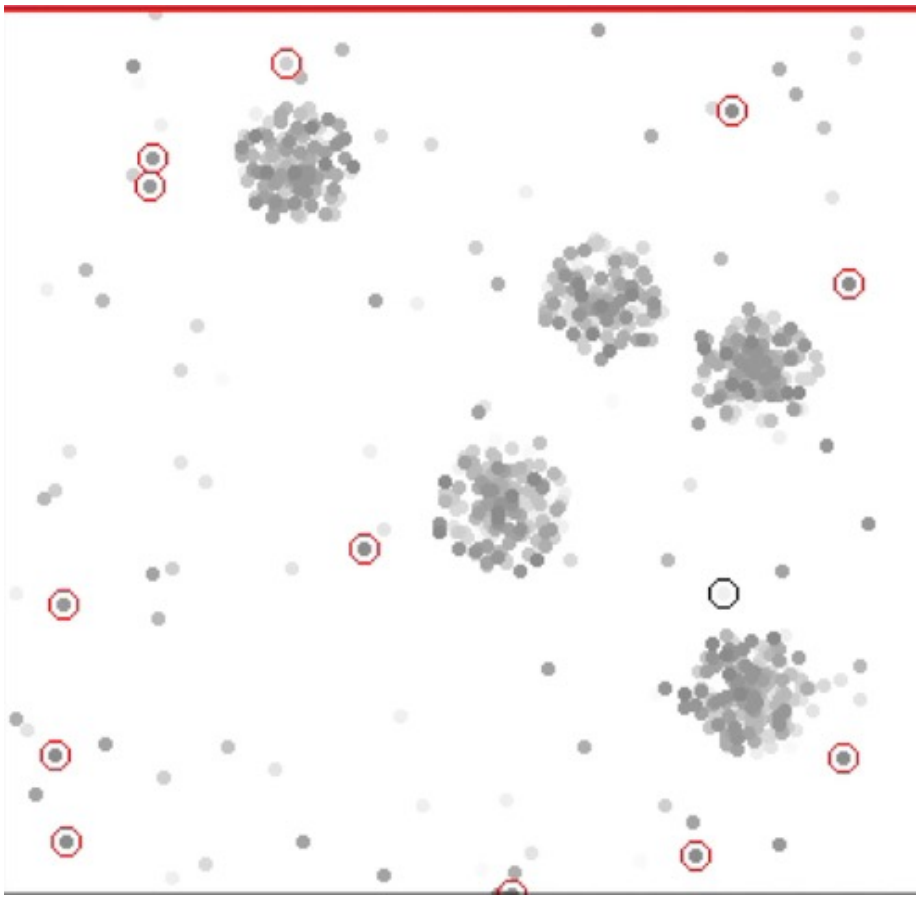
Anomaly Detection

- Build profiles from normal behavior of users, networks, hosts, etc.
- Detection is based on identifying significant deviations from these profiles
- There is no profile for the anomalies



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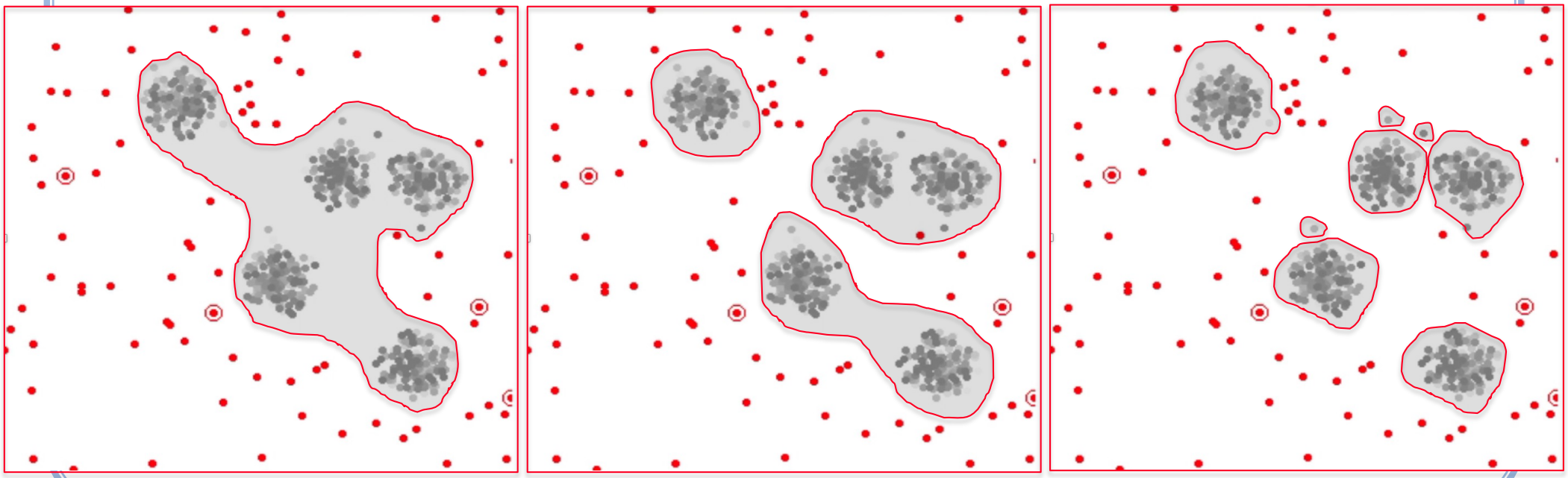
Anomaly Detection

- **The challenge is to define the border between regions**
  - **This border is usually represented by a score threshold**
- **An example:**
  - **Gray points are normal events**
  - **Red points are anomalies**



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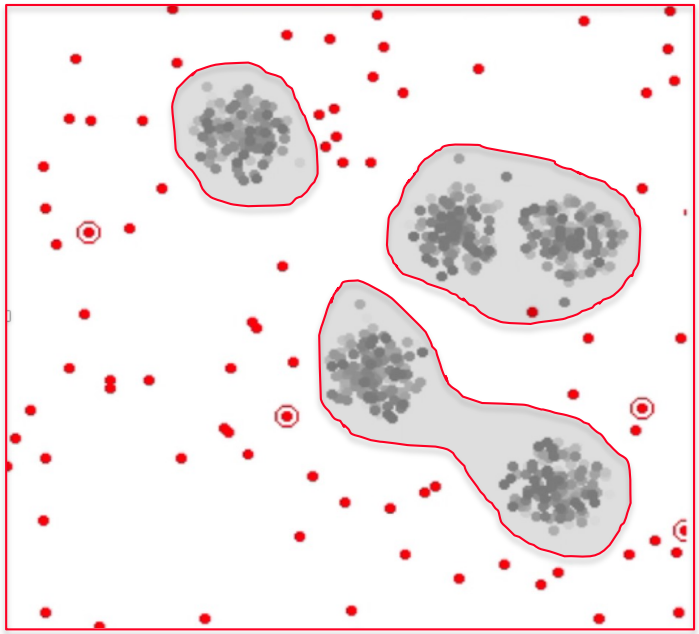
Anomaly Detection

- The borders are usually not clear
- A first step to have a reasonably clear separation is the existence of representative data sets
- This must include:
  - Adversarial behavior (e.g., attacks and compromised machines)
  - Dynamic behavior (behavioral changes along the time)
  - Noisy data



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Anomaly Detection

- **Challenges on Network Intrusion (Anomaly) Detection**
  - **Big data**
    - » **Millions of connections per day/hour in a big organization**
  - **Dimensionality of the data**
    - » **Amount of event attributes is normally large**
  - **Temporal data**
    - » **Order of collected events is relevant**
  - **Unbalanced classes**
    - » **Anomalies in general are very rare**
  - **Real-time detection is desired**
    - » **It's always worth detecting problems earlier**

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Anomaly Detection "Meta-Techniques"
(i.e., machine learning techniques)

- **Supervised: there is a list of pairs of input (e.g., event such as flow report) and output (e.g., normal or attack label), and the algorithm tries to learn the mapping between inputs and outputs**
  - **E.g., classifiers**
- **Unsupervised: There is no mapping, so the algorithm just try to find regularities in the inputs (e.g., flows), i.e., the inputs follow certain patterns and some patterns (e.g., normal behavior) are more common than others. We want to see what generally happens and what does not**
  - **E.g., clustering algorithms**
- **Semi-supervised: Parts of the input is mapped, and other parts are not. E.g., there are training data for the normal behavior**
  - **This is classical anomaly detection!**
- ***Obs: In this lecture we focus on supervised techniques as they usually provide better results***

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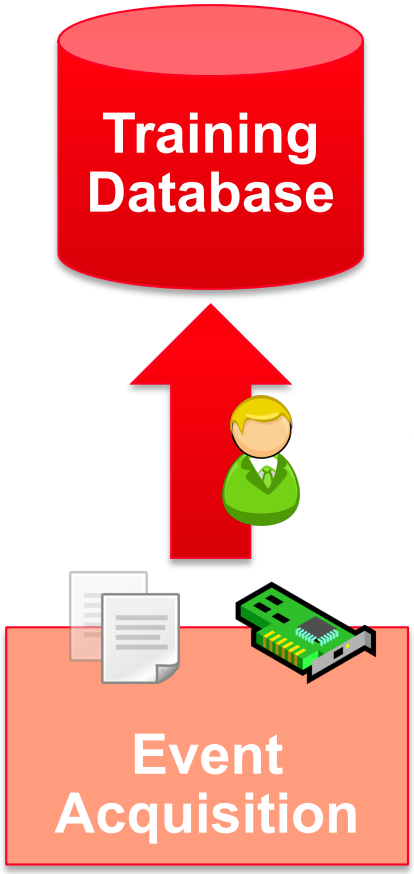
Building an Anomaly Detector



- The environment is monitored, and a training database is created
- All events on the database are classified (i.e., assigned a label)
  - E.g., NORMAL or ATTACK
- Classification is done either:
  - Manually by a specialist
  - Signature-based IDS
  - Clustering
  - ...

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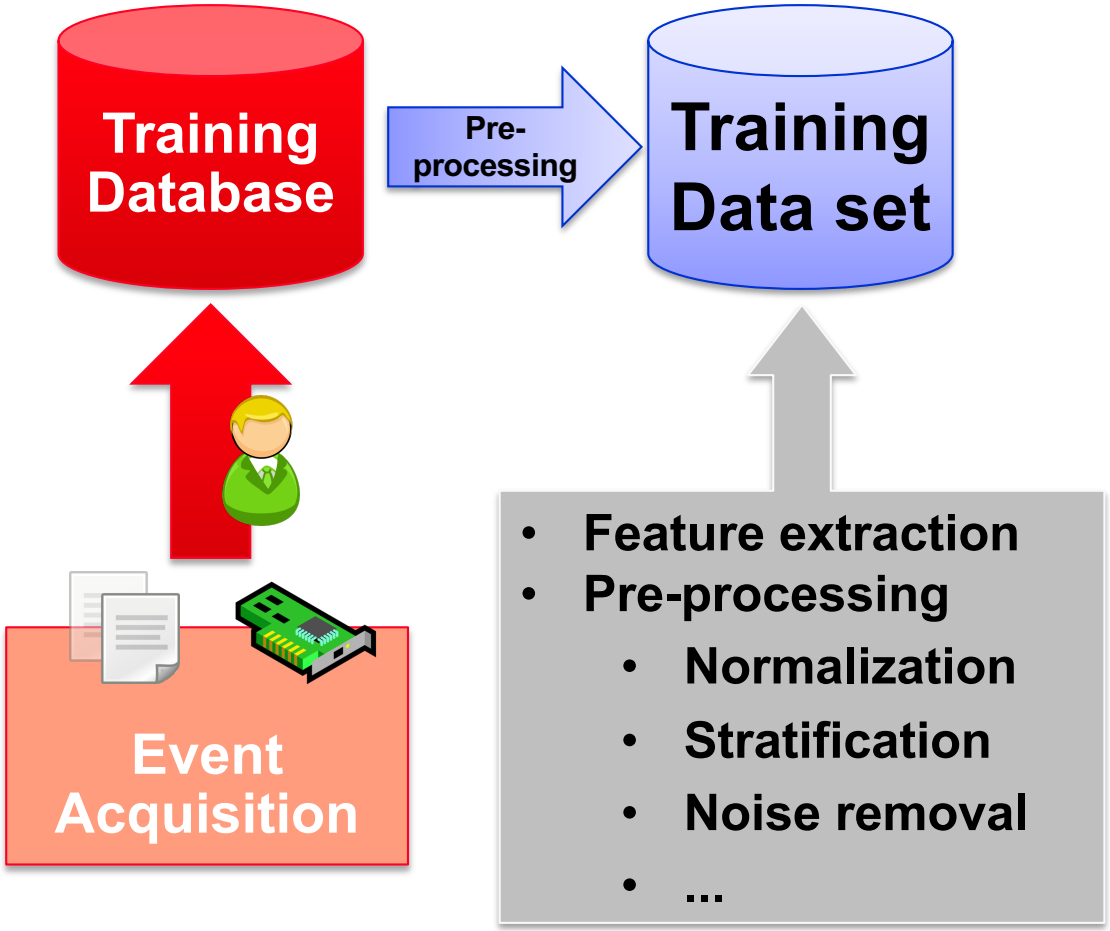
Building an Anomaly Detector



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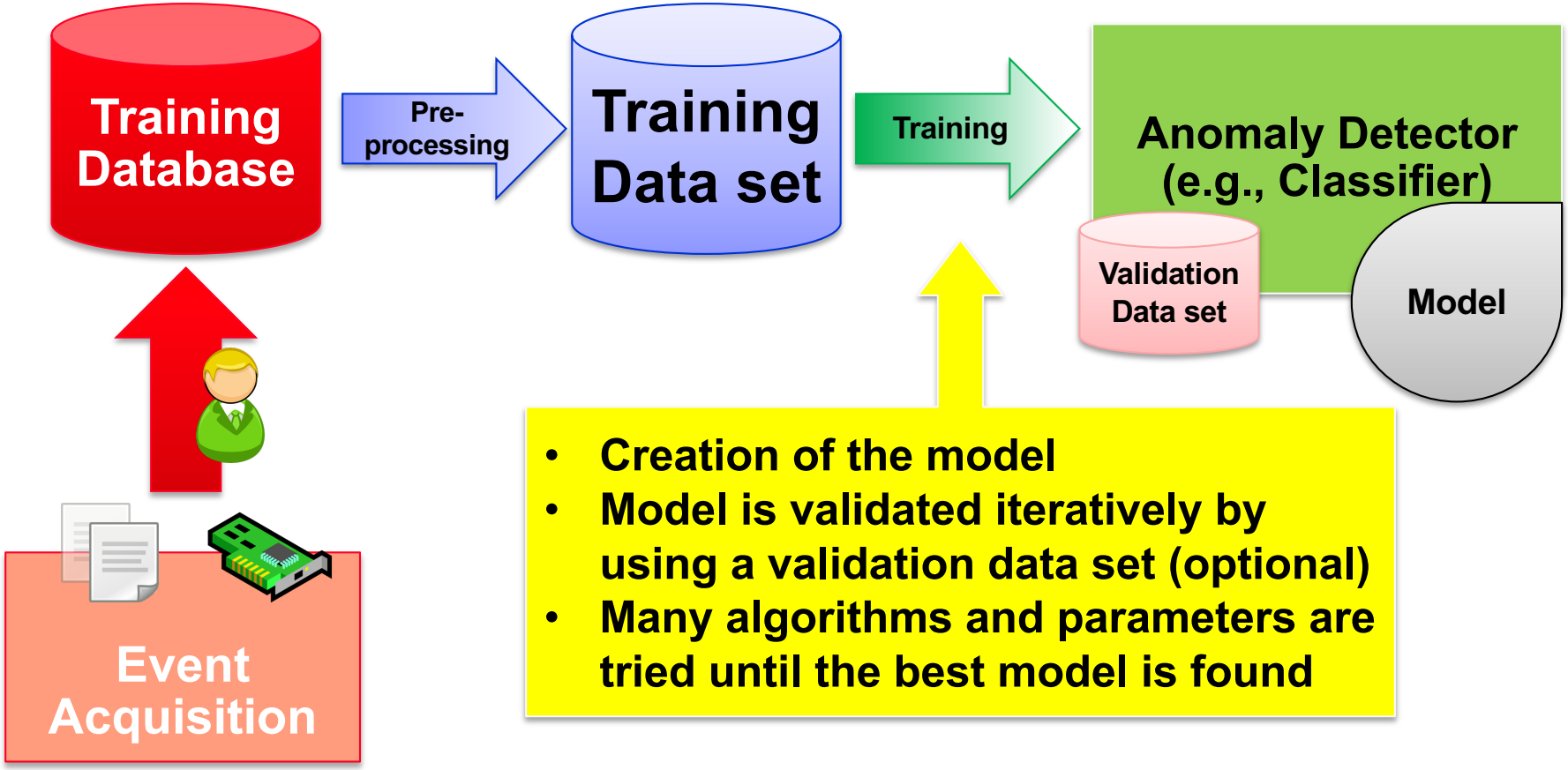
Building an Anomaly Detector



- Creation of the model
- Model is validated iteratively by using a validation data set (optional)
- Many algorithms and parameters are tried until the best model is found

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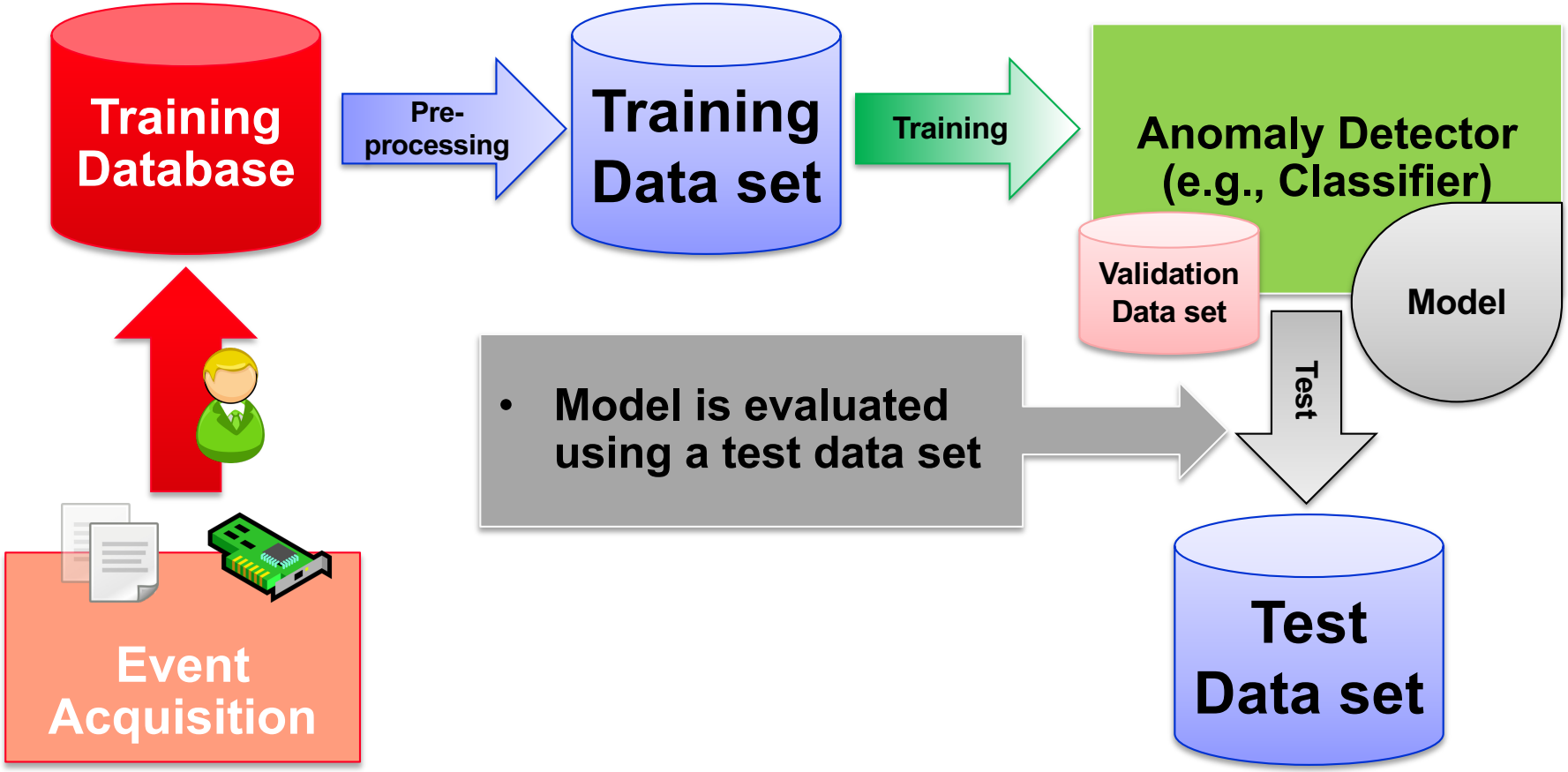
Building an Anomaly Detector



- Model is evaluated using a test data set

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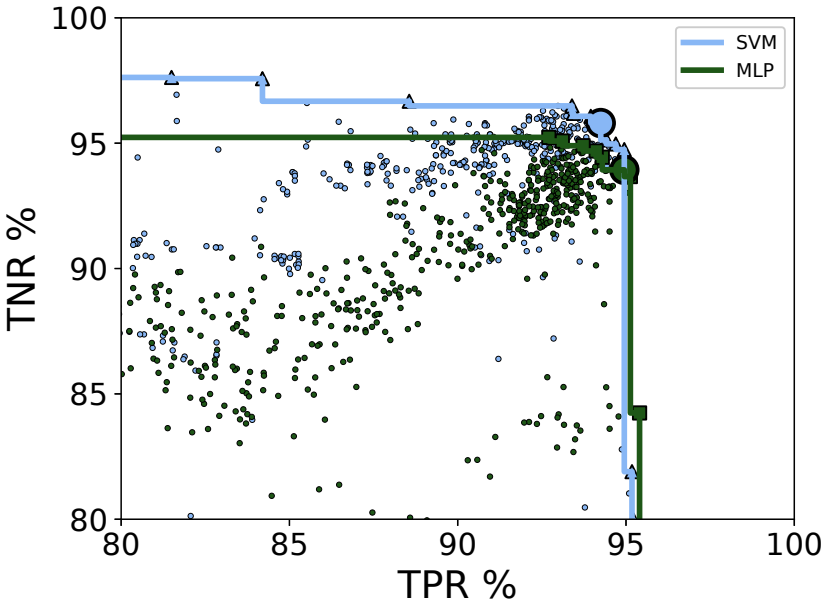
Building an Anomaly Detector
(Example)

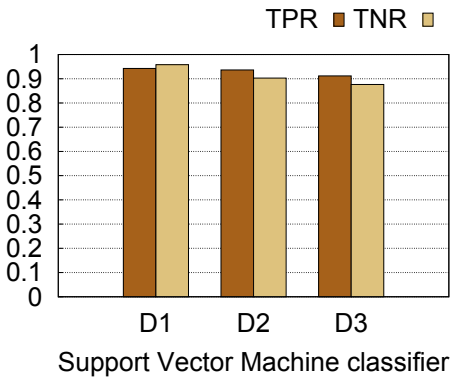
Three datasets (**D1**, **D2** and **D3**) are used for training and testing two classifiers.

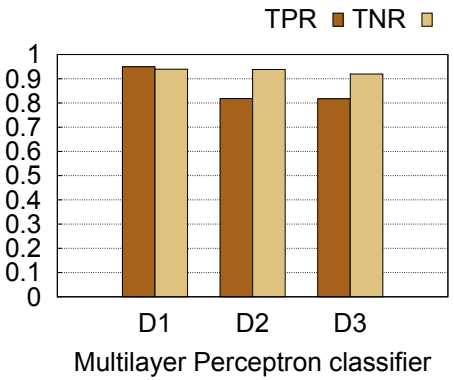
Finding the best model



Uses dataset **D1** for training and validation of a **SVM** (Support Vector Machine) and a **MLP** (Multi-Layer Perceptron): the best models are typically the Pareto optimal (big dots)

Testing the model





Datasets **D2** and **D3** are used for testing the best models (results for D1 are also presented as a reference). Notice:

- Results for D2 and D3 are worse than results for D1
- Results for SVM and MLP are quite different

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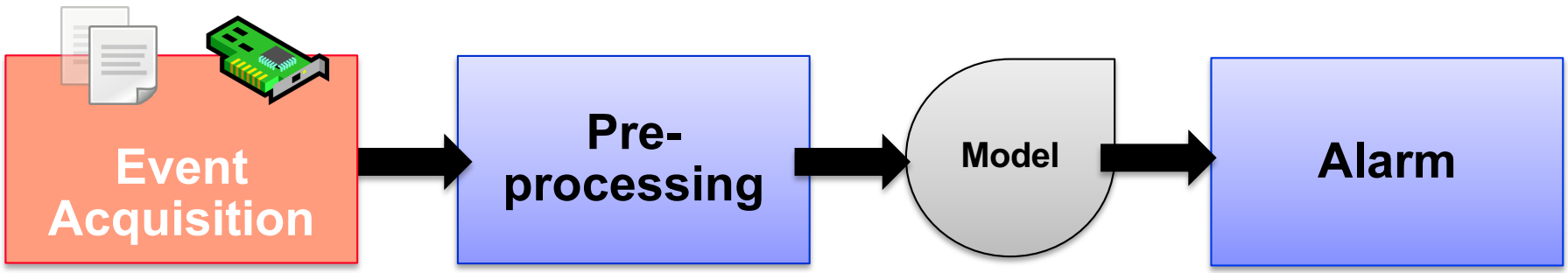
Anomaly Detectors in Production



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Pre-processing Data Sets

- **Data collected in the system must be pre-processed before being used for training, validation and testing**
- **Pre-processing datasets:**
  - **Assign labels such as NORMAL, ATTACK 1, ATTACK 2, etc.**
  - **In some cases, values must be normalized (e.g., between -1 and 1)**
  - **Carefully remove strange values (noise, outliers)**
  - **…**
- **Training and validation data sets typically can be divided in a proportion of 2 to 1, e.g.:**
  - **Training is done using 2/3 of the collected events**
  - **Validation is done using the other 1/3**
  - **(usually this is done several times, for different random subsets to validate the robustness of the anomaly detector)**

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Features

- Machine learning algorithms take as input one or more lists of events (i.e., the observations)
- Each event contains several attributes or features
  - E.g., a flow contains the data volume, duration, starting time, etc.
- Each algorithm must be configured/trained to classify events as normal, or attack based on a subset of the event features
  - E.g., we can train a classifier to look only to the data volume, duration, and TCP flags of a flow to detect anomalies
- An important question is *"What features should we use?"*
  - The challenge is to implement a classifier using the minimum set of representative features

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Features

- **Why not use all available features?**
  - **The complexity (and performance) of any classifier is directly related with the number of inputs (i.e., features of events)**
    - » Typically, #features = #dimensions of event vectors
  - **Simpler models are more robust on small datasets**
  - **Fewer features make it easy to understand and visualize the classifier behavior**
  - **Some of them does not matter for the problem**
  - **Some features are redundant**
- **The number of features can be reduced using <u>dimensionality reduction</u> techniques:**
  - **Feature selection: select *k* out of *d* features that give us the most information and discard the others**
  - **Feature extraction: find a set of *k* < *d* features that are combinations of the original *d* features**

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Techniques for Anomaly Detection

- **Classification-based**
  - **Neural Networks**
  - **Bayesian Networks**
  - **Support Vector Machines**
  - **Rule-based**
- **Nearest neighbor-based**
  - **Distance to the *k*-th Nearest Neighbor**
  - **Relative Density**
- **Clustering-based**
- **Statistical**
  - **Parametric**
  - **Nonparametric**
- **Information-theoretical**
- **Spectral**

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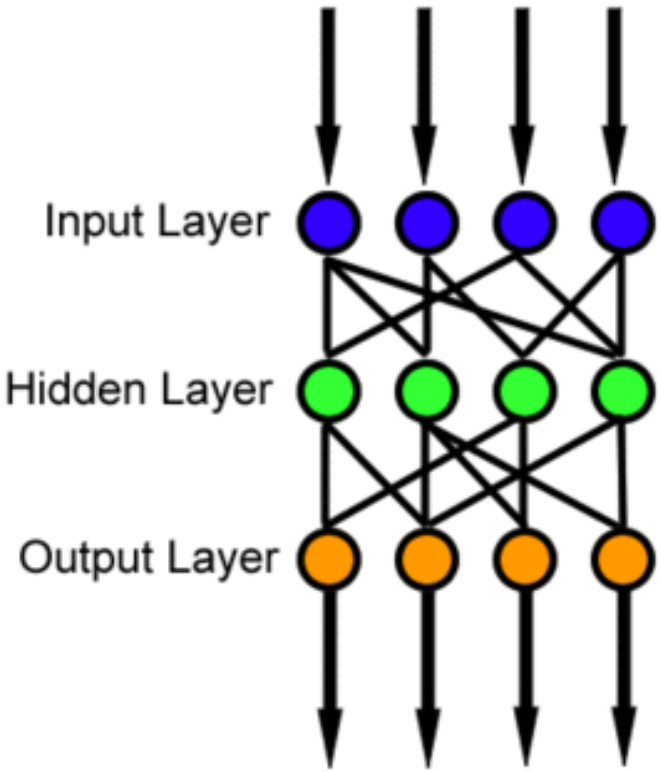
Neural Networks (Multi-Layer Perceptron)

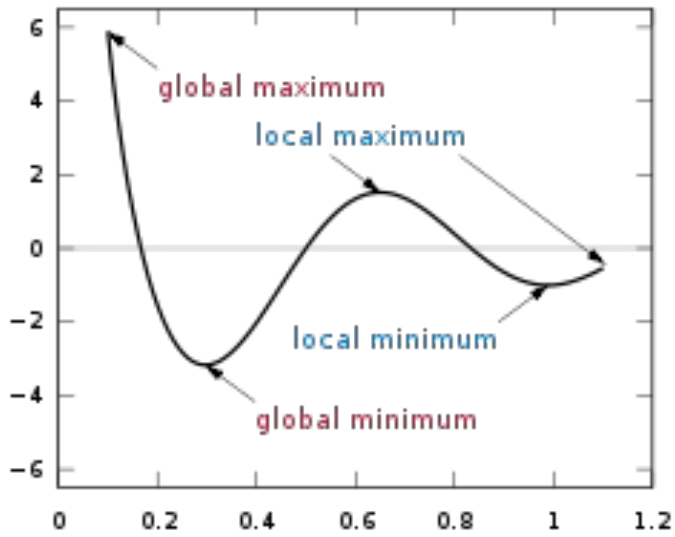
Input **event** with selected features



MLP is a general function interpolator

Class of the event (e.g., NORMAL or ATTACK)

Training is done using the gradient descent algorithm



Images from wikipedia

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k^{th} Nearest Neighbor (*k*NN)

- **One of the simplest machine learning algorithms**
  - **Supervised: need to be trained using annotated datasets**
  - **Non-parametric: data does not need to follow a statistical distribution**
  - **Lazy learning: all computation is deferred until classification**
- **How the (basic) algorithm works?**
  - **Training:**
    - » **Labelled data (e.g., NORMAL, ATTACK) is stored**
  - **Classification of an event *e*:**
    - » **Find the *k* closest neighbors of *e***
    - » ***e* receives the classification corresponding to the majority of these neighbors, e.g.,**
      - **If more than *k*/2 nearest neighbors are NORMAL, the event is normal**
      - **Otherwise, the event is ATTACK**
- **Variation: assign a number of votes (weight) v_n proportional to the distance between the neighbor *n* and *e***

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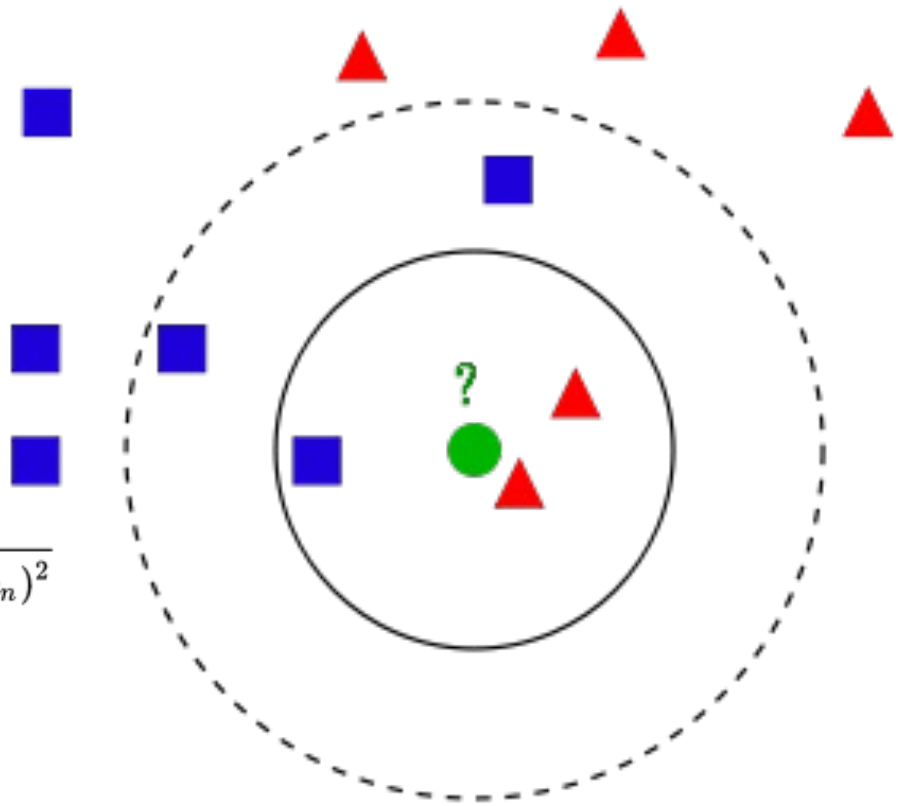
k^{th} Nearest Neighbor (*k*NN)

- **How distances between vectors are calculated?**
- **Typically, the Euclidian Distance is used:**
  - **The distance between $p=(p_1,\ldots,p_n)$ and $q=(q_1,\ldots,q_n)$ in an *n*-dimensional space is:**

$$\mathrm{d}(\mathbf{p},\mathbf{q}) = \mathrm{d}(\mathbf{q},\mathbf{p}) = \sqrt{(q_1-p_1)^2+(q_2-p_2)^2+\cdots+(q_n-p_n)^2}$$

$$= \sqrt{\sum_{i=1}^{n}(q_i-p_i)^2}.$$

- **Other distances are possible**



https://en.wikipedia.org/wiki/K-nearest_neighbors_algorithm#/media/File:KnnClassification.svg

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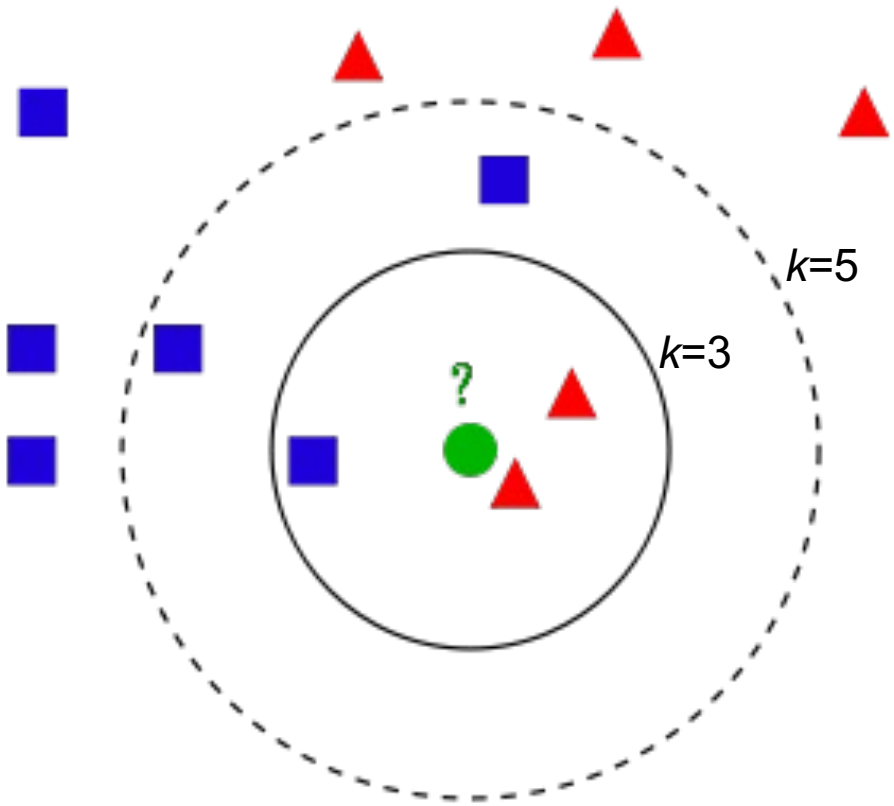
k^{th} Nearest Neighbor (kNN)

- **An example:**
  - **Let triangles be ANOMALY and squares be NORMAL**
  - **The circle is a new event to be classified**
- **If *k=3* the event will be classified as ANOMALY**
- **If *k=5* the event will be classified as NORMAL**
- **What if k=1?**



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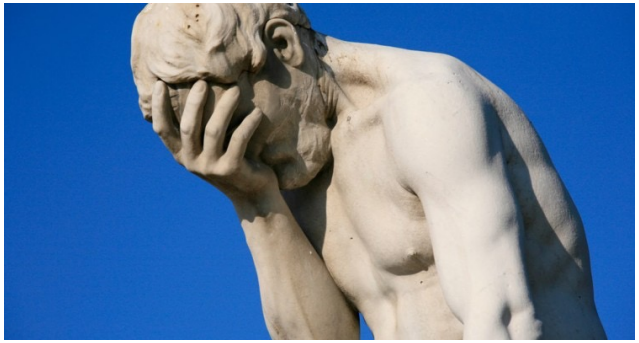
Anomaly-based IDS: Reality Check

- **"Anomaly detection techniques allow the detection of new attacks (zero-day)"**
  - ***Research literature, circa 1999***
- **Machine learning techniques are usually employed for anomaly-based intrusion detection**
  - **However, semi-supervised (traditional) anomaly detectors could solve the problem (partially), but they tend to have a high error rate (in particular, false positives)!**
  - **Classifiers (supervised) perform well but require examples from both normal and attack classes!**
  - **Consequently, a classifier cannot be trained with unknown events…**
  - **<u>So, it is not "expectable" that a classifier will detect new attacks</u>**



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Anomaly-based IDS: Reality Check

- Nonetheless, ML-based anom. detectors are still useful:
  - ML algorithms identify known patterns
    - » Typically, events by all classes are needed, however
    - » It is not possible to generate all known attacks
    - » It is not possible to generate all variations of known attacks
    - » It is not possible to generate unknown attacks
  - A good anomaly-based IDS can detect:
    - » Known attacks (saw during training)
    - » Similar attacks (variations of known attacks)
    - » New attacks (never saw before, but sufficiently different from normal behavior)
- Can we do that with signature-based IDSes?

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Consider these Attacks

Category	Attack Type	Tool Used	Description
DoS	SYNFlood	Hping3	Sends several requests to open TCP connections varying the attack send frequency and the duration time
	UDPFlood	Hping3	Sends several UDP datagrams to an open DNS port varying the attack send frequency, the duration time and the size of each datagram
	ICMPFlood	Hping3	Sends several ICMP messages to the target varying the attack send frequency, the duration time and the size of each datagram
	TCP Keepalive	Slowloris	Initiate several HTTP connections and keep them open for a period of time, varying the number of connections to be open
	SMTPFlood	Postal	Sends several emails to an SMTP server varying the duration time, body size, subject size and frequency
	HTTPFlood	LOIC	Sends several HTTP-get requests to an specific URL, varying the duration time and frequency
Probe	UDPScan	Nmap	Searches for open UDP ports varying the attack send frequency and the duration time
	SYNScan	Nmap	Searches for open TCP ports by sending TCP packets with the SYN flag set while varying the attack send frequency and the duration time
	NULLScan	Nmap	Searches for open TCP ports by sending TCP packets without flags set while varying the attack send frequency and the duration time
	TCPConnect	Nmap	Searches for open TCP ports by completing the three-way handshake while varying the attack send frequency and the duration time
	FINScan	Nmap	Searches for open TCP ports by sending TCP packets with the FIN flag set while varying the attack send frequency and the duration time
	XMASScan	Nmap	Searches for open TCP ports by sending TCP packets with the FIN, PSH and URG flags set while varying the attack send frequency and the duration time
	ACKScan	Nmap	Searches for open TCP ports by sending TCP packets with the ACK flag set while varying the attack send frequency and the duration time
	OS Fingerprint	Nmap	Identify the OS from the target (https://nmap.org/book/osdetect.html) while varying the attack send frequency and the duration time
	Service Fingerprint	Nmap	Identify the services and their versions from the target (https://nmap.org/book/man-version-detection.html) while varying the attack send frequency and the duration time
All	Vulnerability Scan	Nessus	Identify service level vulnerabilities while varying the attack send frequency and the duration time

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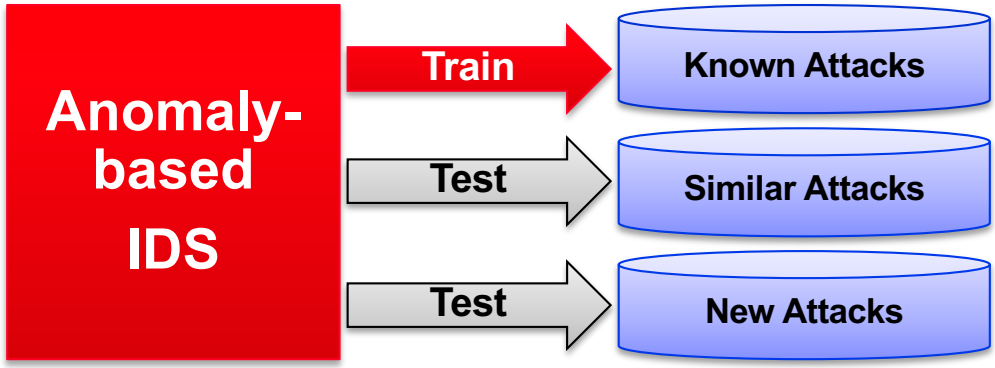
A sound method…

- **Example**

Scenario	Attacks Generated	Traffic (Network Packets)			Size (Megabytes)
		Background	Attack	Total	
Probe (Known)	UDPScan, SYNScan, NULLScan, TCPConnect, FINScan, XMASScan and ACKScan	28,618,365	36,628	28,654,993	8.476
Probe (Similar)	OS Fingerprint and Service Fingerprint	28,477,884	10,441	28,488,325	8.499
Probe (New)	Vulnerability Scan	28,391,914	17,753	28,409,667	8.512
DoS (Known)	SYNFlood, UDPFlood, ICMPFlood and TCP Keepalive	26,747,521	761,269	27,508,790	7.945
DoS (Similar)	SMTPFlood and HTTPFlood	40,278,594	26,390,723	66,669,317	12.143
DoS (New)	Vulnerability Scan	27,522,317	3,429	27,525,746	7.265

  - **Method (notice that the same applies to normal behavior)**



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In Summary

- **Signature-based IDS (e.g., Snort, Suricata) are simple**
  - **Detect attacks and intrusions based on simple comparisons**
  - **Do not detect new attacks**
  - **May be difficult to use with many signatures**
  - **But it works (no false positives) and is effective in many cases!**
- **Anomaly-based IDS**
  - **Estimated error-rate usually does not match what happens in production environments**
  - **Careful training-validation-test process can help in building a decent anomaly detector**
  - **There will be mistakes:**
    - » **False negatives: people can live with few of these (signature-based IDSes have many!)**
    - » **False positives: this need to be improved (recall the base rate fallacy), either by aggregating alarms or using an ensemble of detectors**

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References

- V. Chandola, A. Banerjee, V. Kumar. **Anomaly Detection: A survey.** ACM Computing Surveys. Jul 2009.
- E. Alpaydin. **Introduction to Machine Learning.** 3rd Edition. The MIT Press. 2014.
- Slides from Eduardo K. Viegas (PUC-PR)

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